

DEEP PATEL

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SUMMARY

AI Engineer with one year of experience as a Machine Learning Engineer and an M.S. in Computer Science. Worked on Python-based projects involving RAG, model fine-tuning, AI agents, vector search, and reranking. Used FastAPI to build backend APIs and worked with OpenAI and Claude models. Also gained experience with AWS and evaluating AI workflows for accuracy, cost, latency, and response quality.

EDUCATION

University of Oklahoma Aug 2023 - May 2025

MS in Computer Science

Coursework: Natural Language Processing, Artificial Intelligence, Machine Learning

Indian Institute of Information Technology (IIIT), India Aug 2018 – May 2022

BTech in Computer Science and Engineering

TECHNICAL SKILLS

Languages: Python, PyTorch

RAG & Retrieval: Qdrant, FAISS, Semantic Search, BM25, Hybrid Search, Metadata Filtering, Reranking, Citation-Grounded Generation

Backend & Data: FastAPI, Pydantic, REST APIs, PostgreSQL, SQLite, Redis

LLM & Agent Systems: Agentic AI, Agentic RAG, Multi-Agent Systems, Agent Orchestration, Tool Calling, Structured Outputs, MCP, Context & Memory Management, Human-in-the-Loop

Evaluation & Observability: LLM Evaluation, MLflow, LLM-as-a-Judge, Retrieval Metrics, Token and Cost Tracking, Latency Tracking, Trace Logging

Cloud & DevOps: AWS, Docker, Git

RESEARCH PROJECTS

Research on Claim Verification and Hallucination Reduction [\[GitHub\]](#) June 2026 – Present

- Researched why **RAG** systems can still produce unsupported answers, even when web search, ranking, and reranking are used.
- Built a prototype called **CAGE (Claim-Aware Grounding and Evaluation)** to test whether generated claims were supported by the retrieved sources.
- Collected evidence from source documents, preserved citation metadata, and used the retrieved content to generate the first draft.
- Used a workflow inspired by Google's **SAFE methodology** to split the draft into self-contained factual claims and verify them using NLI (Natural Language Inference), citation matching, and an LLM-based evaluator.
- Marked each claim as supported, unsupported, or irrelevant, then removed unsupported statements before returning the final answer.

EXPERIENCE

Machine Learning Engineer May 2022 – Jul 2023

Firenix Technologies

- Worked on image classification, face recognition, and anomaly detection workflows for security applications using **Python and PyTorch**.
- Used a **YOLO**-formatted image dataset as the primary dataset to train convolutional neural network models.
- Applied k-fold cross-validation during training and monitored the loss values to compare model performance.
- Created a separate test dataset after training. Since the number of available images was limited, cleaned and resized the images and applied data augmentation to improve the dataset.
- Achieved 85% accuracy and reduced false security alerts by 20%. Used **MLflow** to track experiments, manage model versions, and compare the performance of different models.

PROJECTS

Agentic AI Investigation Platform

[\[Website\]](#) Mar 2026 – Jun 2026

- Worked on the **AI orchestration** part of a blockchain investigation platform. The system creates investigation reports, transaction timelines, and fund-flow graphs from the collected data.
- Designed the agent flow to understand the investigation request, plan the next steps, call the required tools, and prepare the final report.
- Connected Moralis and Etherscan APIs to retrieve wallet transactions, token transfers, counterparties, and multi-hop fund movements.
- Used **GPT-5 mini** for agent reasoning and **GPT-5 nano** for checking the generated results. Added session-based context so users could continue earlier investigations.
- Added tool-call tracing, latency tracking, **LLM evaluation**, evidence checking, and **LLM-as-a-Judge** to find unsupported answers and failures in the investigation flow.
- Added report export in PDF, DOCX, and JSON formats. During testing, the system achieved around 80% answer quality and 90% evidence coverage.

Research Document Question-Answering Platform

[\[GitHub\]](#) Aug 2025 – Dec 2025

- Built a question-answering platform where users can upload up to 10 research documents and ask questions based on their content. The answers are returned with citations from the uploaded files.
- Cleaned the uploaded documents, divided the text into smaller chunks, generated embeddings using **all-MiniLM-L6-v2**, and stored them in a vector database along with the document metadata.
- Combined **FAISS** semantic search and **BM25** keyword search for hybrid retrieval. Used a **Cross-Encoder reranking model** to reorder the retrieved chunks and select the most relevant context.
- Evaluated the retrieval pipeline using **Recall@K** and **MRR**, achieving approximately 80% retrieval accuracy.
- Added **session-based memory** to keep track of earlier questions and help with follow-up queries. **OpenAI SDK** was used for response generation, with **Ollama** added as a local fallback.

Efficient Fine-Tuning of GPT-2 and BERT

[\[GitHub\]](#) Feb 2025 – May 2025

- Fine-tuned **GPT-2** and **BERT-base** using **LoRA** and **QLoRA** to study parameter-efficient alternatives to full model fine-tuning.
- Used **bitsandbytes** for 4-bit quantization, with LoRA rank ($r=8$) for GPT-2 and ($r=32$) for BERT-base.
- Fine-tuned GPT-2 on the **XSum** dataset and evaluated it using perplexity. Fine-tuned BERT-base on the **SST-5** dataset and evaluated it using classification accuracy.
- QLoRA reduced GPT-2 perplexity by **2.75** compared with the base model and improved BERT classification accuracy by **0.14%**. The experiments were run on an **NVIDIA L40S GPU**.